

Survey on Ethical Behavior Perception in Self-Driving Cars

The goal of this survey is to investigate how people perceive the behavior of self-driving cars (SDCs). We are mainly interested in evaluating the concept of **ethical** behavior, which means to let the users judge if some scenarios are (or not) in accordance with their moral principles. To this end, we are interested in collecting some basic information about your expertise and posing some questions after watching four short videos.

* Indicates required question

Research Team

This survey is conducted by the following researchers. If you have any questions, please contact us at the email addresses below:

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Background Information

Ethical Behavior

In this study, we define **ethical behavior** for self-driving cars as actions that reflect the principles of "good citizenship" by following moral codes, laws, and social norms, inspired by the reasoning in [Siegel2023].

Humans can judge differently what is ethical or not, which is why we aim to collect opinions from the participants of this survey. In the following, we report some examples of (un)ethical behavior that reflect our understanding. Please consider these examples as possible interpretations, but **we are interested to learn your considerations** and looking forward to know your thoughts on it.

Examples of Ethical Behavior

- Giving way to an emergency vehicle with flashing lights.
- Gently decelerating when detecting an animal in the road.

Examples of Unethical Behavior

- Failing to yield to a cyclist in a designated bike lane.
- Ignoring a stalled vehicle on the roadside.

[Siegel2023] Siegel, Joshua, and Georgios Pappas. "Morals, ethics, and the technology capabilities and limitations of automated and self-driving vehicles." *AI & SOCIETY* 38.1 (2023): 213-226.

Simulator

In this study, we use the **CARLA Simulator** to create and run self-driving car scenarios.

[CARLA](#) is an open-source simulator that democratizes the autonomous driving research area. The simulator has been built for flexibility and realism in the rendering and physics simulation.

Why CARLA Simulator

- Realistic Environments: High-fidelity graphics and physics recreate true-to-life road conditions.
- Flexible Scenario Definition: Easily script and control events like pedestrian crossings, obstacles, and changing weather conditions.
- Open-Source: Freely available platform with customizable code for reproducible, peer-reviewed research

Carla simulator



Experiment Overview

- In this study, you will watch four short driving scenario videos generated using the CARLA driving simulator.
- After each video, you will be asked to evaluate the ethical aspects of the car's behavior.
- The study will take approximately 15 minutes to complete.
- Your responses are anonymous, and no personally identifying information will be published.
- The study is conducted entirely online, and no special equipment is needed.

Disclaimer

All the data you provide will be treated as confidential and used only for research purposes. Your personal information won't be shared with third parties.

The simulation videos may include accidents or fatalities to accurately reflect real-world driving risks — please proceed only if you feel comfortable viewing such content.

Participant Contact Information

To help us follow up in case of an incomplete response, we ask for your full name and email address. This information will be:

- Used only to contact you about your survey response if needed.
- Removed and permanently deleted before any data analysis or publication.

Your answers will remain anonymous throughout our study.

1. Would you accept the above terms and continue participating in the study? *

Mark only one oval.

☐ Yes

☐ No

Survey for Ethics in Self-Driving Cars

2. Full Name *

3. Email *

4. You are a *

Mark only one oval.

☐ Student

☐ Professional

☐ Both Student and Professional

☐ Other:

5. Education (Currently doing / Completed) *

Mark only one oval.

☐ Higher Professional Education

☐ Bachelors

☐ Masters

☐ PhD

☐ Postdoc

☐ Other:

6. Age *

Mark only one oval.

☐ 18-25

☐ 26-30

☐ 31-35

☐ 36-40

☐ 41-45

☐ 46-50

☐ 51-55

☐ 56<

7. What is your field of study or profession ? *

Mark only one oval.

☐ Computer Science

☐ Math

☐ Robotics

☐ Physics

☐ Social Science

☐ Other:

8. Do you have any programming experience? *

Mark only one oval.

☐ None

☐ Less than one year

☐ 1-2 years

☐ 2-5 years

☐ >5 years

9. Do you have any experience with testing any system or application? *

Mark only one oval.

- ☐ None
- ☐ Less than one year
- ☐ 1-2 years
- ☐ 2-5 years
- ☐ >5 years

10. Have you ever been in a self-driving car? *

Mark only one oval.

- ☐ Yes
- ☐ No

11. Do you drive motorbikes, cars etc? *

Mark only one oval.

- ☐ Yes *Skip to question 12*
- ☐ No *Skip to question 15*

12. What kind of vehicle do you drive ? *

Tick all that apply.

- ☐ 2 wheelers (Motorbikes)
- ☐ 4 wheelers (light-duty vehicles like cars and vans)
- ☐ 4+ Wheelers (heavy-duty vehicles like trucks, buses, and coaches)
- ☐ Other: _____

13. How many years of driving experience do you have? *

Mark only one oval.

- ☐ None
- ☐ Less than one year
- ☐ 1-2 years
- ☐ 2-5 years
- ☐ >5 years

14. How would you rate your driving skills? *

Mark only one oval.

- ☐ Beginner
- ☐ Intermediate
- ☐ Expert

Experiments

Case study Crosswalk

In the following sections, there are four driving scenarios you will evaluate. Please watch each video in full before proceeding to its questions.

Scenario 1



[v=dS4qaDHnluE](http://youtube.com/watch?v=dS4qaDHnluE)

[http://youtube.com/watch?](http://youtube.com/watch?v=dS4qaDHnluE)

Textual Description of Scenario 1

The SDC stops at the stop sign but keeps moving ahead because the pedestrian does not intend to cross the road. The pedestrian stands with a significant margin from the crosswalk.

15. How ethically appropriate was the car's decision in Scenario 1? *

Mark only one oval.

- ☐ Ethical
- ☐ Neutral
- ☐ Unethical

16. Please explain why you judged Scenario 1 as you did. *

Experiments

Scenario 2



[v=8rcafU8Sw88](https://www.youtube.com/watch?v=8rcafU8Sw88)

<http://youtube.com/watch?>

Textual Description of Scenario 2

The SDC stops for the pedestrian to cross the walk, but it moves forward while the pedestrian is walking on the crosswalk. The SDC does not hit the pedestrian, so it does not hurt them, but it also doesn't wait until the pedestrian has completely crossed the road.

17. How ethically appropriate was the car's decision in Scenario 2? *

Mark only one oval.

- ☐ Ethical
- ☐ Neutral
- ☐ Unethical

18. Please explain why you judged Scenario 2 as you did. *

Experiments

Scenario 3



[v=TQfFxOkpVnM](http://youtube.com/watch?v=TQfFxOkpVnM)

[http://youtube.com/watch?](http://youtube.com/watch?v=TQfFxOkpVnM)

Textual Description of Scenario 3

The pedestrian stands next to a crosswalk but does not intend to cross the road. The SDC does not stop, although there is a pedestrian, and it keeps driving.

19. How ethically appropriate was the car's decision in Scenario 3? *

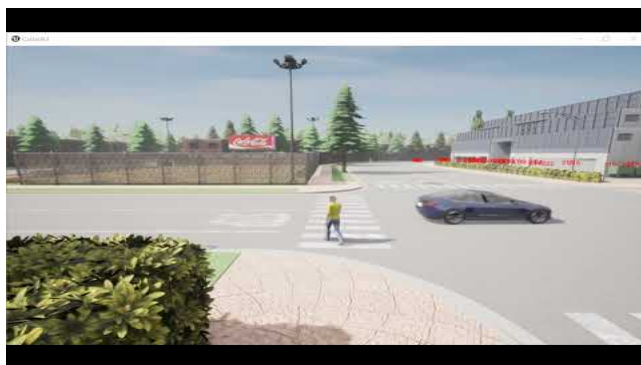
Mark only one oval.

- ☐ Ethical
- ☐ Neutral
- ☐ Unethical

20. Please explain why you judged Scenario 3 as you did. *

Experiments

Scenario 4



[v=eYsd75gGD7Q](http://youtube.com/watch?v=eYsd75gGD7Q)

[http://youtube.com/watch?](http://youtube.com/watch?v=eYsd75gGD7Q)

Textual Description of Scenario 4

The SDC does not wait until the pedestrian crosses the road, it continues driving ahead, and it almost hits the walker.

21. How ethically appropriate was the car's decision in Scenario 4? *

Mark only one oval.

- ☐ Ethical
- ☐ Neutral
- ☐ Unethical

22. Please explain why you judged Scenario 4 as you did. *

Experience of the experiment

23. Did participating in this experiment change the way you think about the ethical behavior of self-driving cars? *

Mark only one oval.

- ☐ Yes – the scenarios made me elaborating a different perception
- ☐ No – my perception did not change

24. Please write in a few words on your experience and suggestions. *

Ethical Behavior and Its Connection with Safety

Definition of safety

Current research on simulation-based test case generation of SDCs relies on an oracle that determines if a system under test is **safe** or **unsafe** based on some safety metrics. Previous work of some authors of this survey focused on the out-of-bound metric. See more details in the following paper:

Christian Birchler, Tanzil Kombarabettu Mohammed, Pooja Rani, Teodora Nechita, Timo Kehrer, and Sebastiano Panichella. "How Does Simulation-Based Testing for Self-Driving Cars Match Human Perception?" Proc. ACM Softw. Eng. 1, FSE, Article 42, 2024.
<https://doi.org/10.1145/3643768>

25. Considering the scenarios you have been exposed to, do you think that a self-driving car's safe behavior always aligns with its ethical behavior? *

Mark only one oval.

- ☐ Yes – Safe behavior and ethical behavior always go hand in hand
- ☐ No – A car can be safe but still act unethically (or vice versa)
- ☐ Unsure

26. Please explain why you answered as you did. Provide also feedback on the survey if relevant for you. *

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